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SUB-SOIL INVESTIGATION REPORT

FOR PROPOSED CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING
AT 38, JALALABAD R/A, SADAR, SYLHET
DAG # 332,333, KHA# -- , J.L # 92, MOUJA- AMBORKHANA

CLIENT

MONAYEM KHAN

Verification Code: SGp1P9xr
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET.

SAMPLE COLLECTED BY

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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-45 ft	09
02	Disturbed sample	02	0-40 ft	08
03	Disturbed sample	03	0-40 ft	08
04	Disturbed sample	04	0-40 ft	08
05	Disturbed sample	05	0-40 ft	08
06	-	-	-	-
07	-	-	-	-
08	-	-	-	-
09	-	-	-	-
10	-	-	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

<u>Bearing capacity for deep foundation</u>

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N;$ $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	19	2.227	2.342	2.554	2.630	2.842	3.091	2.400
	5.0-10	24	2.756	2.899	3.160	3.255	3.516	3.825	2.970
BH02	2.5-5.0	21	2.413	2.538	2.766	2.850	3.078	3.349	2.600
	5.0-10	21	2.384	2.507	2.733	2.816	3.042	3.309	2.569
BH03	2.5-5.0	9	1.090	1.147	1.250	1.288	1.391	1.513	1.175
	5.0-10	8	0.940	0.989	1.078	1.110	1.199	1.305	1.013
BH04	2.5-5.0	22	2.529	2.660	2.899	2.987	3.226	3.510	2.725
	5.0-10	20	2.328	2.448	2.669	2.749	2.970	3.231	2.509
BH05	2.5-5.0	16	1.879	1.976	2.155	2.219	2.398	2.608	2.025
	5.0-10	19	2.212	2.326	2.536	2.612	2.822	3.070	2.384
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

Table 8: The allowable skin friction for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.049	0.192	0.072	0.024	0.024	-	-	-	-	-
10	0.168	0.192	0.120	0.144	0.136	-	-	-	-	-
15	0.208	0.128	0.032	0.280	0.176	-	-	-	-	-
20	0.192	0.176	0.048	0.056	0.144	-	-	-	-	-
25	0.096	0.248	0.168	0.056	0.152	-	-	-	-	-
30	0.216	0.232	0.232	0.232	0.184	-	-	-	-	-
35	0.224	0.224	0.224	0.224	0.224	-	-	-	-	-
40	0.216	0.216	0.216	0.216	0.216	-	-	-	-	-
45	0.208	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.450	38.400	14.400	4.800	4.800	-	-	-	-	-
10	33.600	38.400	24.000	28.800	27.200	-	-	-	-	-
15	41.600	25.600	6.400	56.000	35.200	-	-	-	-	-
20	38.400	35.200	9.600	11.200	28.800	-	-	-	-	-
25	19.200	49.600	33.600	11.200	30.400	-	-	-	-	-
30	43.200	46.400	46.400	46.400	36.800	-	-	-	-	-
35	44.800	44.800	44.800	44.800	44.800	-	-	-	-	-
40	43.200	43.200	43.200	43.200	43.200	-	-	-	-	-
45	41.600	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.032	0.125	0.047	0.016	0.016	-	-	-	-	-
10	0.109	0.125	0.078	0.094	0.088	-	-	-	-	-
15	0.135	0.083	0.021	0.182	0.114	-	-	-	-	-
20	0.125	0.114	0.031	0.036	0.094	-	-	-	-	-
25	0.062	0.161	0.109	0.036	0.099	-	-	-	-	-
30	0.140	0.151	0.151	0.151	0.120	-	-	-	-	-
35	0.146	0.146	0.146	0.146	0.146	-	-	-	-	-
40	0.140	0.140	0.140	0.140	0.140	-	-	-	-	-
45	0.135	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.203	17.280	6.480	2.160	2.160	-	-	-	-	-
10	15.120	17.280	10.800	12.960	12.240	-	-	-	-	-
15	18.720	11.520	2.880	25.200	15.840	-	-	-	-	-
20	17.280	15.840	4.320	5.040	12.960	-	-	-	-	-
25	8.640	22.320	15.120	5.040	13.680	-	-	-	-	-
30	19.440	20.880	20.880	20.880	16.560	-	-	-	-	-
35	20.160	20.160	20.160	20.160	20.160	-	-	-	-	-
40	19.440	19.440	19.440	19.440	19.440	-	-	-	-	-
45	18.720	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner, following recommendations are suggested for the proposed six (06) storied residential building.

Soil profile scenario

The overall soil information of the investigated site is quite irregular among the boreholes, and the soil formation is notable for its vertical inconsistency. Almost two types of borehole scenarios have been observed.

For type A (borehole no.: 01, & 05)- gray soft clay and brown sandy clay with poor bearing capacity exist roughly up to the top 5 ft depth below EGL, followed by various color fine sand with good bearing capacity up to 20 ft depth. Although gray sandy clay having moderate bearing capacity has been found at 25 ft depth below EGL in borehole no.:01, it did not continue deeper. Finally, brown fine sand with good bearing capacity initiates in those boreholes at 30 ft of EGL and continues thereafter.

For type B (borehole no.: 02, 03 & 04)- gray fine sand with good bearing capacity exists roughly up to the top 10 ft from EGL in all boreholes. After that, brown and gray sandy clay layers with low bearing capacity have been observed from 15-25 ft depth below EGL in another two boreholes except borehole no.:02. Finally, a brown fine sand layer with good bearing capacity initiates from 25 ft of EGL in those boreholes and continues to a greater depth.

From the above discussion it is very clear that the whole scenario of the proposed site is quite different in nature.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Shallow foundations may not be a suitable option for six storied residential building as different color sandy clay soil with low bearing capacity has already been observed from 15-25 ft depth of EGL. Although, bearing capacity values for shallow foundations can be found from the attached **Table 7** in the report which is based on existing soil conditions.

However, deep foundation may be preferentially recommended to ensure the stability of the structure. Size and depth of the pile should be selected in the light of bearing capacity given in the report.

Safety provisions

Since the proposed site is composed of alternate layers of soft cohesive and granular soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **settlement** of the site considering building load while designing either shallow or deep foundation. Obviously, **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. Client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **45** ft as client provided.

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